

PREPARE Challenge - Phase 2: Explainability Report

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1. Abstract

We explored how best to communicate predictions to a broad audience of lay preclinical patients, their care providers and clinicians. On the technical side, we use SHAP (Shapley Additive Explanations) values to explain how predictions are made at the individual level. To provide deeper understanding, we also consider and compare the rigor of three techniques for estimating prediction intervals and opt for a conservative approach based on Conformal Prediction.

Our approach to targeting is to combine AI best practice with Design Thinking, to provide information that is clear, accurate and easy for non-technical people to understand. We reviewed best practice literature on the subject and took feedback from a range of patients and healthcare professionals.¹ We learned:

- Most people understand that their actual score is unlikely to be exactly the same as their prediction - but what is the likely range, and how likely is it? We answer this using prediction intervals, which (like confidence intervals) estimate a range for a score with a certain probability. Choosing round numbers, we settle on 60% (the true score is more likely to be within this range than outside it), and 90% prediction intervals.
- Range is more useful than decline, as decline predictions are not yet reliable enough.
- To add context, we visualize the predicted score and likely range against the overall population mean and distribution, using color to reflect risk with green, amber and red.
- We avoid technical terms such as RMSE and MAE to describe the accuracy of the model.
- We provide text alongside the figures and charts, explaining in accessible language how to interpret them, limitations, and where to go for further help, to mitigate potential harms and aid people from different backgrounds in understanding the information.

Due to space limitations, the explainers are missing some information that clinicians and lay readers have said they would like to see. They would like more detail about how predictions are made, confidence, how ranges are calculated and what the limitations are. They would also like to know how the prediction relates to actual risk of AD/ADRD, for which we could explore prediction models using diagnosis as a label. We wonder if the reports should extend to several pages. We could also include predictions from different models under different scenarios – it would be interesting to explore how lifestyle changes can affect outcomes.

We choose the following three individuals for our example explainers:

- 'fnfh' as a typically aging example (2016 score 234, 2021 score 242)
- 'yeeg' as a cognitive decline example (2016 score 148, 2021 score 103)
- 'jepa' as an example where our approach predicts and handles a very low score very effectively (2016 score 72, 2021 score 38)

2. Technical Approach

How a prediction is made

Of the three approaches we used to explore feature importance at the global level, SHAP Analysis has the benefit of being able to explain most effectively how individual predictions are made.² We also evaluated LIME but consider its output to be very hard for lay people to interpret.³

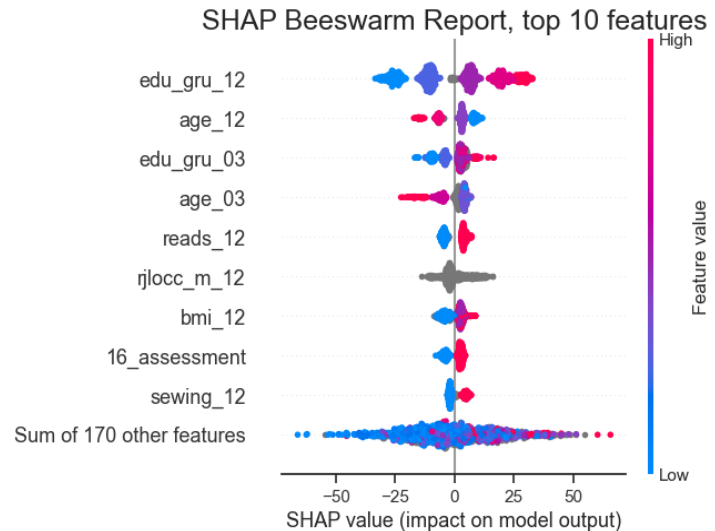
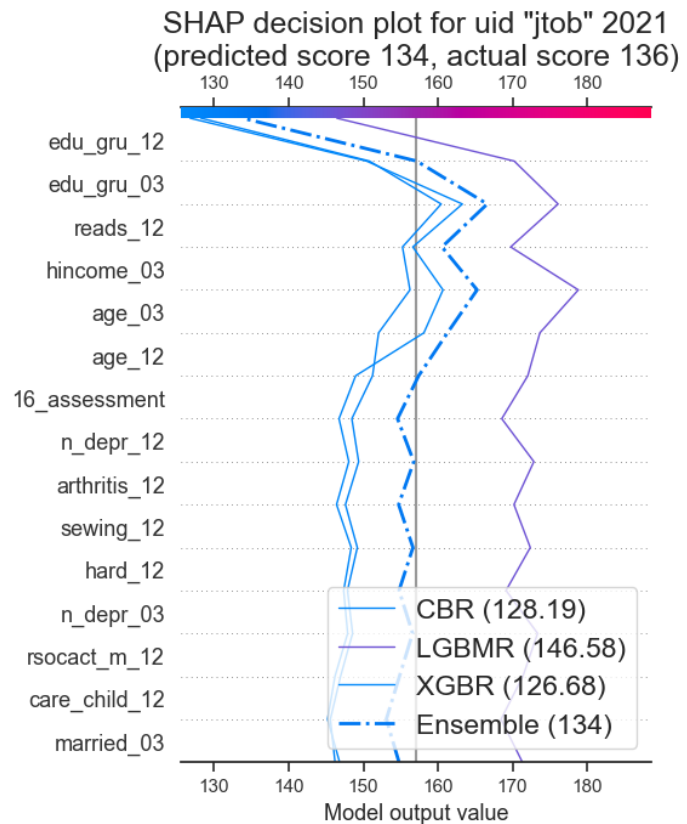
The SHAP Decision Plot on the right focuses on a single prediction. It shows the most important features for that prediction, and how they each quantitatively contribute to it. There is a line for each of the three models in the ensemble, and a dotted line for the overall prediction. Starting at the bottom, a line moves to the left when a feature value decreases the prediction, and to the right when it increases it, arriving at the final output values at the top.

The SHAP Beeswarm Report shows, at a global level, how the values of features affect different predictions to different degrees. Blue represents low values, pink represents high values, and the x-axis represents the impact of that feature value on individual predictions.

Prediction Intervals

Prediction intervals are necessarily wider than confidence intervals, as we need to account for the uncertainty of the individual observation. We compared three methods to estimate intervals for each prediction, with 60% and 90% likelihoods:

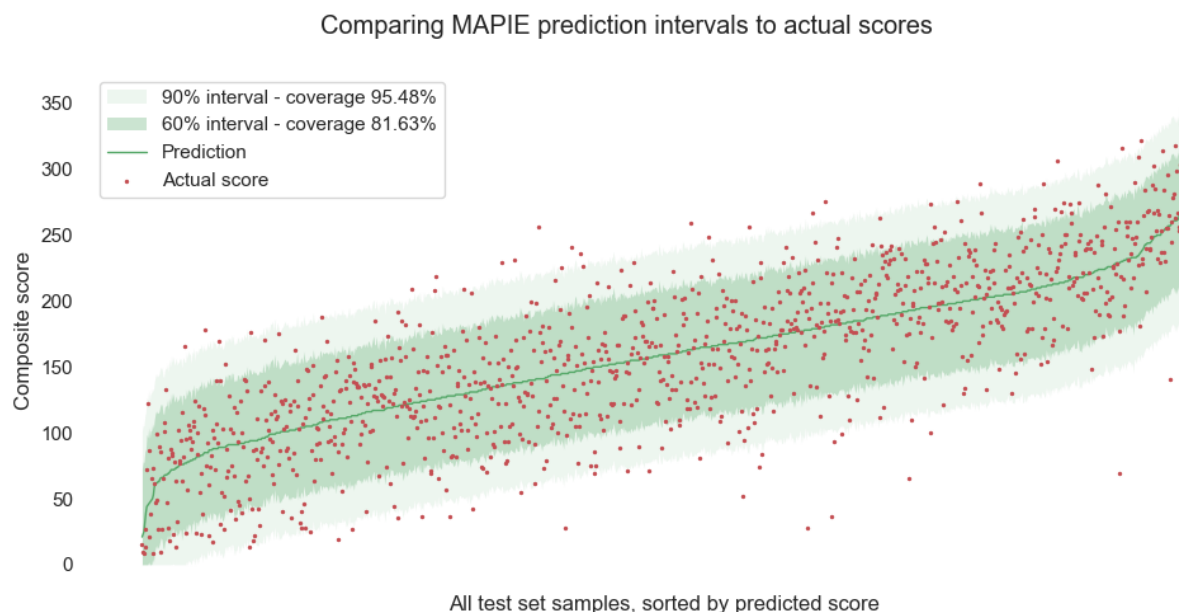
1. Quantile regression for each of the three sub-models, following a method set out by Christopher Hansen of IBM, and taking the mean or outer values of the three for each prediction.⁴ This gave mean intervals of 68 and 161, with coverage of 58% and 89%, on the



test set, for the 60% and 90% intervals respectively. It is observed to be less accurate for very low and high predicted and actual scores.

2. Conformal Prediction using MAPIE (Model Agnostic Prediction Interval Estimator), following an approach set out by Molnar.⁵ This gave wider mean intervals of 102 and 158, with 82% and 95% coverage respectively, and was more reliable across the range.
3. We trained our main model to predict the squared-error for each prediction, as an estimator for its variance - an approach set out by Qucit.⁶ Assuming error is distributed normally, this can then be used to calculate intervals. This provided the narrowest mean intervals of 67 and 131, with 62% and 90% coverage respectively, but was also less accurate for very low and high scores.

Further work is needed on these approaches; for the present we settle on the more conservative MAPIE model for the explainers. The chart below compares predictions and their MAPIE prediction intervals with actual score for each sample in the test set. In the individual explainers, we describe the prediction intervals in the text and show them in a chart in relation to the overall population distribution to help readers with context and intuition on what they mean.



Other metrics

To add further meaning, we include a risk score of high, medium or low in each explainer. The details are to be agreed with clinical experts, but for the present our approach is to apply 'high' to any score over one standard deviation below the mean, 'medium' to any score less than one standard deviation below the mean, and 'low' to any score above the mean. Finally, we provide non-technical figures for overall model accuracy. We multiply RMSE by the z-score 1.645 to estimate a 90% confidence interval. This works out at ± 63 , and we observe that on the test set 89.95% of true scores fall within this range of the prediction. We use this to communicate accuracy intuitively.

3. References

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- ⁴ Casper Hansen, 'Prediction intervals explained: A LightGBM tutorial', at *IBM Developer* (07 April 2022), <https://developer.ibm.com/articles/prediction-intervals-explained-a-lightgbm-tutorial/> (accessed 18 January 2025).
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- ⁶ Qucit, 'A simple technique to estimate prediction intervals for any regression model', Medium.com (6 April 2016), <https://medium.com/@qucit/a-simple-technique-to-estimate-prediction-intervals-for-any-regression-model-2dd73f630bcb> (accessed 19 January 2025).